

Parameter tables

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A.1 Default setpoints

Table A.1. System performance defaults (SonixED skid v2025-A7)

Parameter	Value
Flowrate per stack	0.5–10 m ³ /h
System flow per skid	up to 30 m ³ /h
Recovery rate	up to 98%
Operating pressure	0.5–2.5 bar
Temperature range	5–55 °C
Specific energy	0.6– 2.0 kWh/m ³
Skid dimensions (l×b×h)	3040×1340×2020 mm
Incoming line voltage	3×400 V AC
Max. current	25 A
Valve voltage	24 V DC
Control voltage (DC bus)	24 V DC
Dosing SET 1 (min)	4.0 mA / 0 impulses/min
Dosing SET 2 (max)	20.0 mA / 180 impulses/min

Table A.3. SalinBloc™ container throughput by format (source: HydroVolta brochure)

Parameter	20 ft unit	40 ft unit
Flow rate	2–30 m ³ /h	15–60 m ³ /h
Approx. daily throughput (22 h/day operation)	44– 660 m ³ /day	330– 1,320 m ³ /day

Feed range for both formats: brackish groundwater, 1,500–20,000 ppm TDS. TODO: no container weight figure was found in any source reviewed, including the container general-arrangement drawings themselves (their title blocks carry no weight/dimension text field, and any dimension lines are vector geometry, not extractable text — a human must open these drawings visually). Obtain the weight from the container manufacturer or a loaded-unit weighbridge reading before publishing lifting/foundation guidance.

A.2 Site-specific settings

TODO: no site-specific commissioning/setpoint record was found; use this table to capture the as-commissioned values for a specific installation.

Table A.5. Site-specific settings template

Parameter	Value

A.3 Alarm limits

The P&ID (appendix B) shows instrument spans used as default alarm spans at each skid position: pressure 0–2.5 bar, conductivity 0–200 mS/cm, with high/low alarm tags (PSA, CEA) at every position. HydroVolta's ThingsBoard warning-code table (a real, deployed alarm definition list, source: 08_IT/01_Thingsboard/Warning table) gives concrete numeric examples for several alarm classes:

Table A.7. Representative alarm/warning codes (source: HydroVolta ThingsBoard warning table)

Code	Class	Condition
W001	General	HV-Skid not powered
W002	General	Ventilation off
W011–W013	Chemical	Antiscalant / H ₂ SO ₄ / HCl tank empty
W101–W110	Level	Main diluate/concentrate/electrolyte/breaking/client tank level low or high (e.g. diluate below 40 cm or above 90 cm; electrolyte below 20 cm or above 80 cm)
W111–W118	Level	Buffer tank low-low / high level
W201–W214	Electrical	Stack stage 1–7 voltage below or above system specification
W300	Flow	Diluate flow below 500 l/hr
W301	Flow	Concentrate flow below 500 l/hr

TODO: this table only samples the documented code ranges; the full warning table has ~57 defined codes and should be transcribed in full by HydroVolta engineering, together with the exact level/pressure setpoints per tank size, once the standard SalinBloc™ tank configuration (chapters 15 to 17) is finalized — these codes come from a specific pilot project's tank/stage configuration and the numeric thresholds may not transfer directly. See the alarm tag list in chapters 6, 12 and 15 to 17 for additional alarms.